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Ice Printing Artificial Blood Vessels

In the field of biomedical engineering, the artificial replication of organs and internal body structures is a prominent field of specialization. Artificial internal structures are key to help replace malfunctioning or nonfunctional body parts. The tremendous demand for structures such as hearts, kidneys, and livers vastly exceeds the immediate supply of donated organs. Around 6,000 Americans die waiting for a donor every year. Due to the vast demand, biomedical engineers are eagerly working towards the creation of efficient and proper artificial organs. However, one big challenge in the development of lab-grown organs is the creation of blood vessel networks that mimic natural structures, from tiny capillaries to larger arteries.

Superhighways of vessels beneath our skin transport blood and nutrients. These vessels branch out to different body structures to help the body move food and oxygen to and from one's organs. The delicacy and intricacy of these blood vessels are what makes it so hard to make outside of the body. Traditional artificial blood vessels did not mimic the natural design accurately. Recently, however, a new vessel-building project was held at Carnegie Mellon University in Pittsburgh, Pennsylvania. New technologies at this study have now allowed bioengineers to use 3D ice printing to create more accurate blood-vessel structures. Feimo Yang, a graduate student in the labs of Philip LeDuc and Burak Ozdoganlar at Carnegie Mellon University, will present their research at the 68th Biophysical Society Annual Meeting, on February 10th, 2024, in Philadelphia, Pennsylvania.

3D ice printing typically involves using a semi-solid form of water in a machine to create 3D structures. However, something different that these bio engineers used involves adding a stream of liquid form water to a very cold surface. This form of 3D printing uses a semi-liquid waterform to form a layering effect while printing. The layering effect is called freeform and that's what creates smooth structures. The 3D ice printing of blood vessel structures is not used by regular water, rather a specialized liquid called heavy water. Heavy water is a form of water where the hydrogen atoms are replaced by deuterium, which gives the water a higher freezing point and helps create the smooth structure. Deuterium is an isotope of hydrogen. In the typical hydrogen molecule, there is one proton, one electron, and no neutrons. Neutrons have the most weight in elementary particles that make up an atom. Deuterium, on the other hand, is composed of one proton, one electron, and one neutron. Due to the additional particle, deuterium is "heavier" than normal hydrogen.

Then, these 3D-printed ice templates are embedded in a gelatin material, GelMA (methacrylated gelatin). Unlike gelatin, a natural protein derived from collagen, GelMA is the gelatin's amino groups that have been partially chemically modified with methacrylate groups. Methacrylate monomers are characterized by alkene groups (hydrocarbons) that contain carbon-carbon double bonds. This modification makes the GelMA photocrosslinkable when combined with a photoinitiator. Photocrosslinking is a kind of chemical crosslinking triggered by UV light and a photoinitiator allowing the hydrogel to form in its natural place of origin at physiological pH and room temperature without using toxic chemical crosslinkers. GelMA plays an important role in the biomedical field due to its great biocompatibility and tunable physicochemical properties. GelMa hardens under UV light mimicking the bio composition of certain natural structures. After the templates are embedded in GelMa, it is exposed to UV light, which simultaneously causes the ice to melt away as well. The end result of what is left are realistic blood vessel channels made of GelMa. Since the researchers are building these structures in the natural environment of the blood vessels, there are no issues regarding transferring the structures yet. Perhaps implants would be done in later research projects. However, aside from building the canal-like vessel structures, the researchers have also introduced endothelial cells.

Endothelial cells are not exactly blood cells, yet they obtain a crucial part of the blood vessel structures. Endothelial cells line the inner walls of the blood vessels, responsible for regulating exchanges between the bloodstream and surrounding tissues. Signals from these cells organize the growth and development of connective tissue cells formed along the surrounding layers of the blood-vessel wall. Even new blood vessels can be developed from the walls of existing small vessels by the outgrowth of endothelial cells. These cells have the capacity to form hollow capillary tubes even when isolated in culture. These cells were able to survive on GelMa for up to two weeks. In the future, it is believed that they could survive for longer periods of time if cultured accordingly.

Overall, the goals and results of the research done recently include a new method to foster the creation of artificial blood vessels. In addition to being useful for organ transplants, lab grown vessels also introduced the possibility of testing the effects of certain drugs and medications, even using the patient’s own blood cells for specific results. In conclusion, this new and unique approach is a step towards creating complex and life-like accurate blood vessel structures in the future.

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